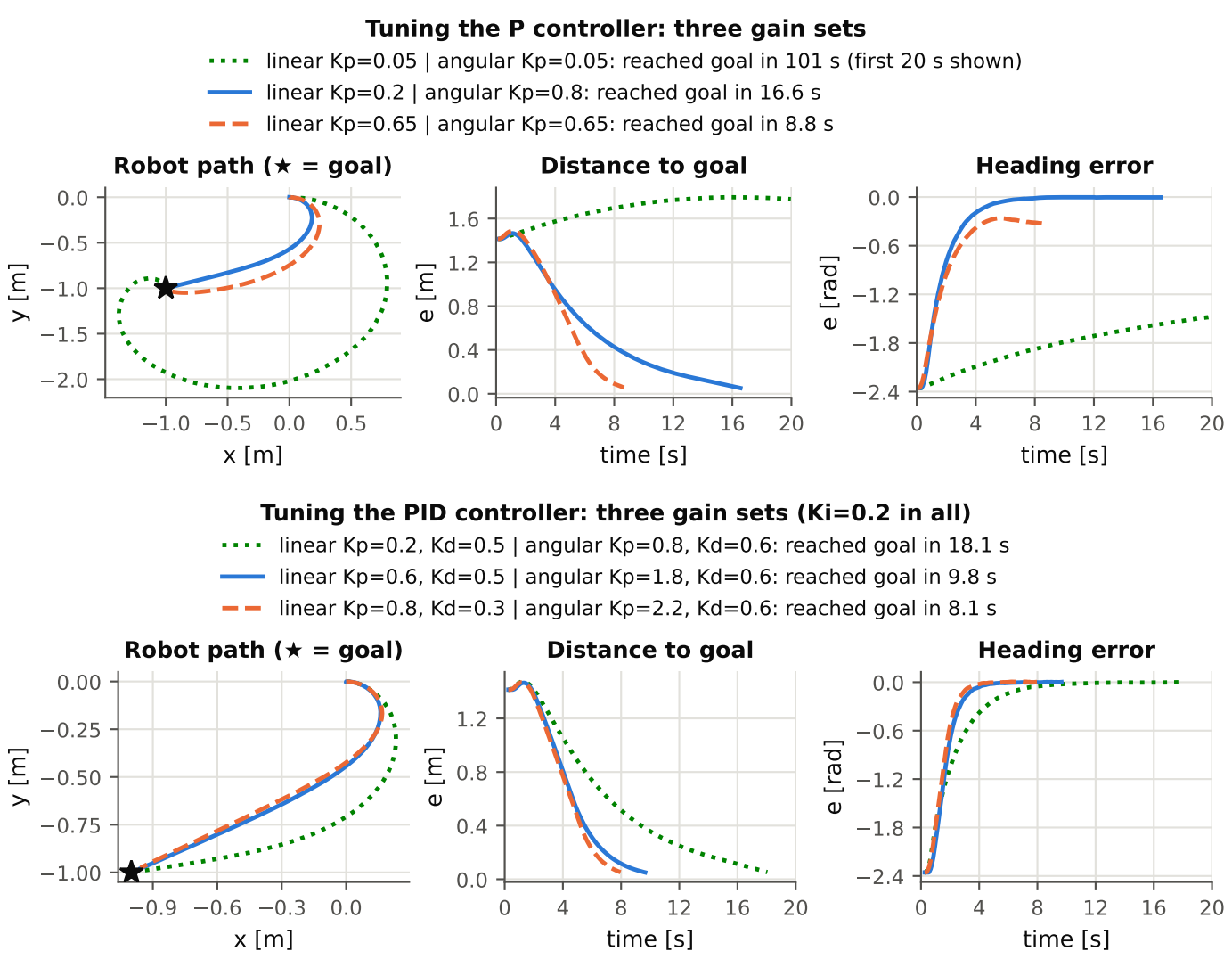
Tuning the P controller: three gain sets
linear Kp=0.05 | angular Kp=0.05: reached goal in 101 s (first 20 s shown)
linear Kp=0.2 | angular Kp=0.8: reached goal in 16.6 s
linear Kp=0.65 | angular Kp=0.65: reached goal in 8.8 s
Robot path (★ = goal)
y [m]
x [m]
Distance to goal
e [m]
time [s]
Heading error
e [rad]
time [s]
Tuning the PID controller: three gain sets (Ki=0.2 in all)
linear Kp=0.2, Kd=0.5 | angular Kp=0.8, Kd=0.6: reached goal in 18.1 s
linear Kp=0.6, Kd=0.5 | angular Kp=1.8, Kd=0.6: reached goal in 9.8 s
linear Kp=0.8, Kd=0.3 | angular Kp=2.2, Kd=0.6: reached goal in 8.1 s
Robot path (★ = goal)
y [m]
x [m]
Distance to goal
e [m]
time [s]
Heading error
e [rad]
time [s]